



Sri Chaitanya IIT Academy., India.

AP, TELANGANA, KARNATAKA, TAMILNADU, MAHARASHTRA, DELHI, RANCHI

A right Choice for the Real Aspirant

ICON Central Office, Madhapur – Hyderabad

Sec: Sr-ICON-B-1

JEE-ADVANCE-2019-P1

Date: 02-05-21

Time: 09:00 AM to 12:00 PM

GTA-02

Max.Marks:186

SYLLABUS:

Physics : TOTAL SYLLABUS

Chemistry : TOTAL SYLLABUS

Mathematics : TOTAL SYLLABUS

Name of the Student: _____

H.T. NO:

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**JEE-ADVANCE-2019-P1-Model**

Time: 3:00Hr's

IMPORTANT INSTRUCTIONS

Max Marks: 186

PHYSICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 1 – 4)	Questions with Single Correct Choice	3	-1	4	12
Sec – II(Q.N : 5 – 12)	Questions with Multiple Correct Choice +1 partial marks	4	-1	8	32
Sec – III(Q.N : 13 – 18)	Questions with Numerical Value Answer Type	3	0	6	18
Total				18	62

CHEMISTRY:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 19 – 22)	Questions with Single Correct Choice	3	-1	4	12
Sec – II(Q.N : 23 – 30)	Questions with Multiple Correct Choice +1 partial marks	4	-1	8	32
Sec – III(Q.N : 31 – 36)	Questions with Numerical Value Answer Type	3	0	6	18
Total				18	62

MATHEMATICS:

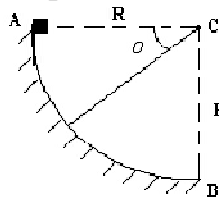
Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 37 – 40)	Questions with Single Correct Choice	3	-1	4	12
Sec – II(Q.N : 41 – 48)	Questions with Multiple Correct Choice +1 partial marks	4	-1	8	32
Sec – III(Q.N : 49 – 54)	Questions with Numerical Value Answer Type	3	0	6	18
Total				18	62

**PHYSICS****Max Marks: 62****SECTION – I**
(SINGLE CORRECT ANSWER TYPE)

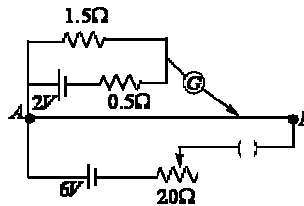
This section contains 4 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which ONLY ONE option can be correct. **Marking scheme: +3 for correct answer, 0 if not attempted and –1 in all other cases.** Section 1 (Max Marks: 12)

- Section 1 contains Four questions
- Each Question has Four Options and Only One of these four will be the correct answer.
- For each question, choose the option corresponding to the correct answer
- The Marking scheme to evaluate Answer to each question will be :
Full Marks: +3 (If the answer is correct)
Zero Marks: 0 (If the question is unanswered)
Negative Marks: -1 (In all other cases)

- 1 A block is released from rest from point A on a smooth track, whose cross section in vertical plane is quarter of a circle of radius 'R'. The value of ' θ ' for which the constant force between block and surface is equal to the weight of block in magnitude is



- A. $\cos^{-1}\left(\frac{2}{3}\right)$ B. $\cos^{-1}\left(\frac{1}{3}\right)$ C. $\sin^{-1}\left(\frac{2}{3}\right)$ D. $\sin^{-1}\left(\frac{1}{3}\right)$
- 2 In fig AB is 300 cm long wire having resistance 10Ω per meter. Rheostat is set at 20Ω . The balance point will be attained at



- A. 1.0m B. 1.25m C. 1.5m D. Cannot be determined
- 3 The equation of a stationary wave is $y = 4 \sin\left(\frac{\pi x}{5}\right) \cos(100\pi t)$. The wave is formed using a string of length 20cm. The second and 3rd antinodes are located at positions (in cm)
- A. 7.5, 12.5 B. 2.5, 7.5 C. 12.5, 17.5 D. 5, 10
- 4 The ratio of cube of circumference of the orbit of a satellite to the volume of the earth is $6 \times 10^{10} (g/R)$, where g is the acceleration due to gravity and R is the radius of earth. Find the time period of satellite (in seconds).
- A. 2×10^5 B. 10^5 C. 4×10^4 D. 10^4



SECTION – II
(ONE OR MORE CORRECT ANSWER TYPE)

This section contains 8 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which ONE OR MORE than ONE option can be correct. **Marking scheme: +4 for all correct options & +1 partial marks, 0 if not attempted and -1 in all wrong cases**

Section 2 (Max Marks: 32)

- Section two contains Eight (8) questions
- Each Question has Four Options and One Or More Than One of these four will be the correct answer.
- For each question, choose the option(s) corresponding to all the correct answer(s)

The Marking scheme to evaluate Answer to each question will be :

Full Marks: +4 (If the answer is correct)

Partial Marks: +3 (If all the four options are correct but **only 3** are chosen)

Partial Marks: +2 (If THREE or More options are correct but **only 2** correct answers are chosen)

Partial Marks: +1 (If TWO or More options are correct but **only 1** correct answer is chosen)

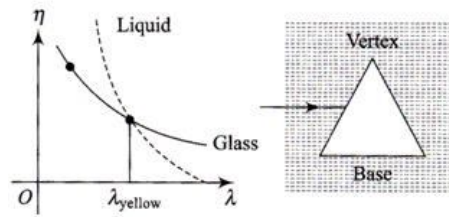
Zero Marks: 0 (If the question is unanswered)

Negative Marks: -1 (In all other cases)

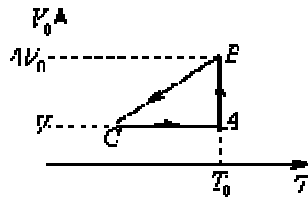
- 5 Two non-conducting solid spheres of radii R and $2R$, having uniform volume charge densities ρ_1 and ρ_2 respectively, touch each other. The net electric field at a distance $2R$ from the centre of the smaller sphere, along the line joining the centres of the spheres, is zero. The ratio $\frac{\rho_1}{\rho_2}$ can be
- A. -4 B. $-\frac{32}{25}$ C. $\frac{32}{25}$ D. 4
- 6 The current density in a wire of radius a varies with radial distance r as $J = kr^2$, where k is a constant.
- A. Total current passing through the cross section of the wire is $I = \frac{\pi ka^4}{2}$
- B. Total current passing through the cross section of the wire is $I = \frac{3\pi ka^3}{2}$
- C. The field at a distance $r > a$ is $B = \frac{\mu_0 ka^4}{4r}$
- D. The field at a distance $r < a$ is $B = \frac{\mu_0 kr^3}{4}$
- 7 For a certain metal, the K absorption edge is at $0.172A^\circ$. The wavelength of K_α, K_β and K_γ lines of K series are $0.210A^\circ, 0.192A^\circ$, and $0.192A^\circ$, respectively. The energies of K, L, and M orbits are E_K, E_L and E_M , respectively. Then $E_K = 13.04keV$
- A. $E_K = -13.04keV$ B. $E_L = -7.52keV$
- C. $E_M = -3.21keV$ D. $E_K = 13.04keV$



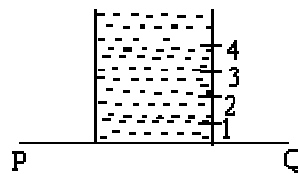
- 8 A glass prism is immersed in a hypothetical liquid. The curves showing the refractive index n as a function of wavelength λ for glass and liquid are as shown in the following figures. When a ray of white light is incident on the prism parallel to the base.



- A. Yellow ray travels without deviation
 B. Blue ray is deviated toward the vertex
 C. Red ray is deviated toward the base
 D. There is no dispersion
- 9 One mole of an ideal gas in initial state A undergoes a cyclic process ABCA, as shown in the Figure. Its pressure at A is P_0 . Choose the correct option (s) from the following



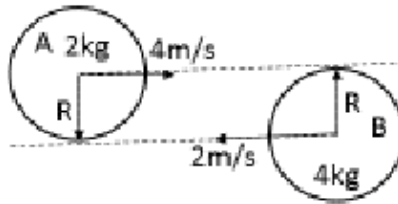
- A. Internal energies at A and B are the same
 B. Work done by the gas in process AB is $P_0 V_0 \ln 4$
 C. Pressure at C is $\frac{P_0}{4}$
 D. Temperature at C is $\frac{T_0}{4}$
- 10 A cylindrical vessel of 90 cm height is kept filled upto the brim. It has four holes 1, 2, 3, 4 which are respectively at heights of 20 cm, 30 cm, 40 cm and 50 cm from the horizontal floor PQ. The water falling at the maximum horizontal distance from the vessel comes from



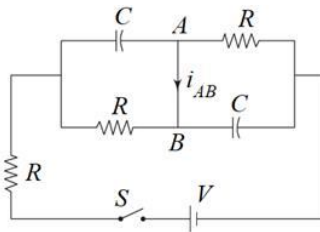
- A. hole number 4 B. hole number 3 C. hole number 2 D. hole number 1



- 11 Two spheres of same radii but mass 2kg and 4 kg are moving toward each other. They collide with restitution coefficient $e=1/3$. During collision the spheres deform and again reform to their original shapes. Then



- A. Magnitude of impulse each sphere experienced during deformation is $8\sqrt{3}Ns$.
 B. Magnitude of impulse during reformation is $4\sqrt{3}Ns$
 C. Loss in kinetic energy of the system during deformation is 18 J.
 D. Loss in kinetic energy of the system during entire collision process is 16 J.
- 12 In the circuit shown capacitors are uncharged initially. S is a switch that is closed at time $t = 0$. Which of the following statements is/are true?



- A. As soon as the switch is closed $i_{AB} = 0$.
 B. At steady state the power dissipated in the circuit is $\frac{V^2}{3R}$.
 C. $i_{AB} = \frac{4V}{3R} \left(e^{-\frac{3t}{RC}} - \frac{1}{4} \right)$
 D. $i_{AB} = \frac{V}{3R} \left(e^{-\frac{3t}{4RC}} - 1 \right)$

SECTION – III (NUMERICAL VALUE TYPE)

This section contains 6 questions. Each question is numerical value type. For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to second decimal place. (e.g. 003.2343, 6.250023, 7.000000, -0.33, -.30, 30.27000, -127.0030).

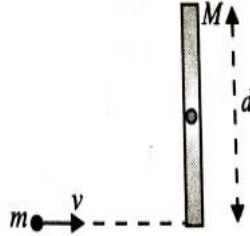
Marking scheme: +3 for correct answer, 0 if not attempted and 0 in all other cases. Section 3 (Max Marks: 18)

- This Section contains **Six (06)** Questions. The answer to each question is a Numerical Value
- For each question, enter the correct numerical value of the answer using the mouse and the on-screen virtual numerical keypad in the place designated to enter the answer. If the numerical value has more than two decimal places, **round off the value to TWO decimal places**.
- The Marking scheme to evaluate Answer to each question will be : Full Marks: +3 (If ONLY the correct numerical value is entered)
Zero Marks: 0 (In all other cases)



- 13 A mass m is moving at speed v perpendicular to a rod of length d and mass $M = 6m$ which pivots around a frictionless axle running through its centre. It strikes and sticks to the end of the rod. Then the angular speed of the system just after the collision is $\frac{v}{xd}$.

Find x



- 14 An inductor of inductance 2.0 mH is connected across a charged capacitor of capacitance 5.0 F and the resulting L - C circuit is set oscillating at its natural frequency. The instantaneous charge on the capacitor is q and the current is I . It is found that the maximum value of q is 200 C . Calculate maximum value of I in amperes.
- 15 An alternating voltage $v(t) = 220 \sin 100\pi t$ volt is applied to a purely resistive load of 50Ω . The time taken for the current to rise from half of the peak value to the peak value is in milliamperes.
- 16 Two nuclei P, Q have equal no. of atoms at $t = 0$. Their half-life is 3 hours, 9 hours. Compare their rates of disintegration after 18 hrs from the start is $\frac{x}{16}$. Find x value
- 17 A screw gauge having 100 equal divisions and a pitch of length 1 mm is used to measure the diameter of a wire of length 5.6 cm . The main scale reading is 1 mm and 47th division on the circular scale is on the reference line. Find the curved surface area of the wire in cm^2 to appropriate significant figures. (use $\pi = \frac{22}{7}$)
- 18 Two rectangular blocks A and B of different metals have same length and same area of cross-section. They are kept in such a way that their cross-sectional area touches each other. The temperature at one end of A is 100°C and that of B at the other end is 0°C . If the ratio of their thermal conductivity is $1:3$, then under steady state, the temperature of the junction in contact will be (in $^\circ\text{C}$) _____

**CHEMISTRY****Max Marks: 62**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 4 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which ONLY ONE option can be correct. **Marking scheme: +3 for correct answer, 0 if not attempted and –1 in all other cases.** Section 1 (Max Marks: 12)

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 Negative Marks: -1 (In all other cases)

- 19 $X + H_2SO_4 \rightarrow$
 Y (a colorless gas with irritating smell)
 $Y + K_2Cr_2O_4 + H_2SO_4 \rightarrow$ green solution
 X and Y are
A. SO_3^{2-}, SO_2 **B.** Cl^-, HCl **C.** S^2, H_2S **D.** CO_3^{2-}, CO_2
- 20 Initial concentration of the reactant is 1.0M. The concentration becomes 0.9M, 0.8M and 0.7M in 2 hours, 4 hours and 6 hours respectively. Then the order of reaction is
A. 2 **B.** 1 **C.** Zero **D.** 3
- 21 The correct order of increasing radius of the elements Si, Al, Na and P is:
A. $Si < Al < P < Na$ **B.** $P < Si < Al < Na$
C. $Al < Si < P < Na$ **D.** $Al < P < Si < Na$
- 22 Among the following configurations, the element which has the highest electron affinity is:
A. $[Ne]3s^13p^2$ **B.** $[Ne]3s^23p^5$ **C.** $[Ne]3s^23p^4$ **D.** $[Ne]3s^23p^63d^54s^1$

SECTION – II
(ONE OR MORE CORRECT ANSWER TYPE)

This section contains 8 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which ONE OR MORE than ONE option can be correct. **Marking scheme: +4 for all correct options & +1 partial marks, 0 if not attempted and -1 in all wrong cases**

Section 2 (Max Marks: 32)

- Section two contains Eight (8) questions
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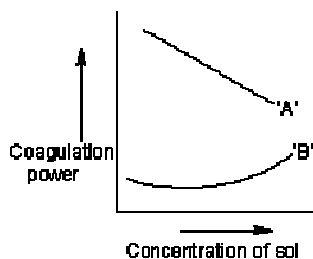
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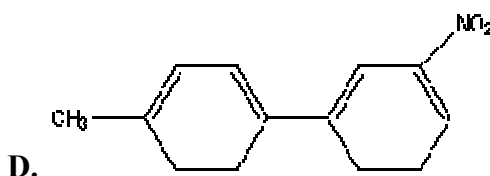
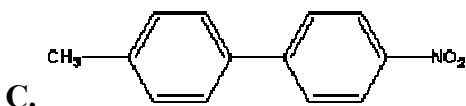
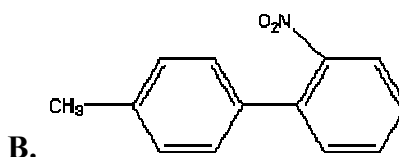
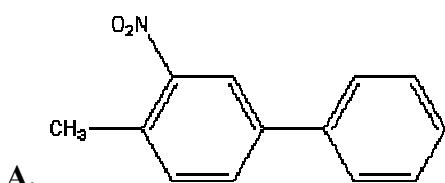
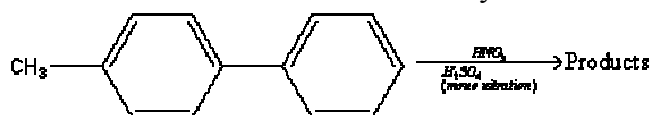


- 23 An unknown substance of formula $M^{+2}X^{-2}$, ($M.W = 144$) crystallizes in one of the cubic lattice, with the density 4.63 g/cm^3 , then the correct options are (edge length of its unit cell $= 5.87 \text{ \AA}$)
- A. $M^{+2}X^{-2}$ crystallizes in fcc manner B. Schottky defect $= 9.8 \times 10^{-2} \text{ gram/cm}^3$
- C. its unit cell consists 8 ions per unit cell D. Coordination number of $M^{+2}X^{-2} = 8:8$
- 24 Following diagram shows the dependence of coagulation value on concentration of sol in case of hydrophobic sol.

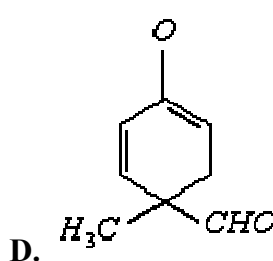
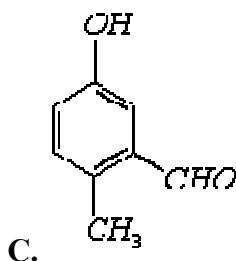
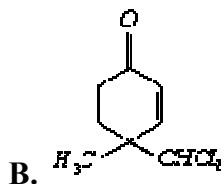
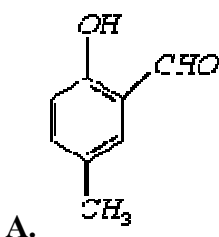


- A. Curve 'A' is for univalent electrolyte B. Curve 'A' is for polyvalent electrolyte
- C. Curve 'B' is for univalent electrolyte D. Curve 'B' is for polyvalent electrolyte

25



- 26 p-cresol, chloroform and caustic potash are heated, the compounds formed are :





27 $R-CH_2-\overset{\overset{O}{\parallel}}{C}-OH \xrightarrow[P/Cl_2]{H_2O} R-\underset{\underset{Cl}{|}}{CH}-\overset{\overset{O}{\parallel}}{C}-OH$ Which of the following are intermediate species responsible for product?

- A. $R-CH_2-\overset{\overset{O}{\parallel}}{C}-Cl$ B. $R-\overset{\cdot}{C}H_2-\overset{\overset{O}{\parallel}}{C}-OH$ C. $R-\underset{\underset{Cl}{|}}{CH}=\overset{\overset{O}{\parallel}}{C}-OH$ D. $R-\underset{\underset{Cl}{|}}{CH}-\overset{\overset{O}{\parallel}}{C}-Cl$

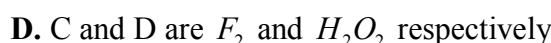
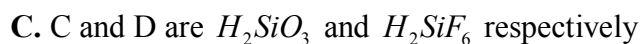
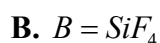
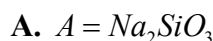
28 In which of the following, the hybrid orbitals of the central atom have the same s -character?

- A. CH_4 B. XeO_3 C. $Ni(CO)_4$ D. $[Ni(CN)_4]^{2-}$

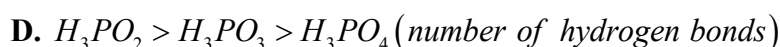
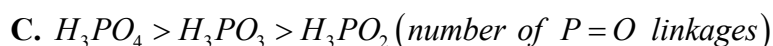
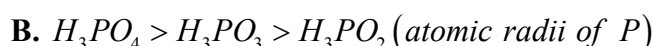
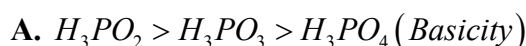


29

Hence



30 Select the incorrect comparison(s) for indicated property:



SECTION – III (NUMERICAL VALUE TYPE)

This section contains 6 questions. Each question is numerical value type. For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to second decimal place. (e.g. 003.2343, 6.250023, 7.000000, -0.33, -.30, 30.27000, -127.0030).

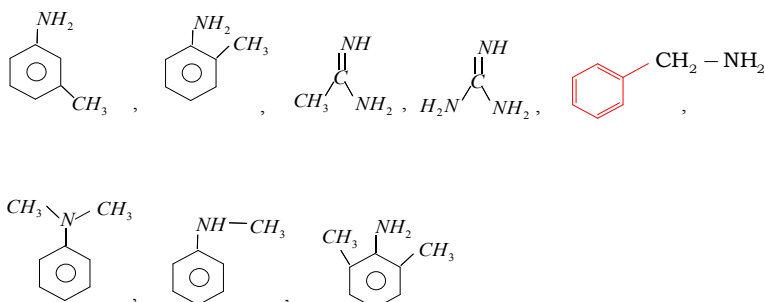
Marking scheme: +3 for correct answer, 0 if not attempted and 0 in all other cases. Section 3 (Max Marks: 18)

- This Section contains Six (06) Questions. The answer to each question is a Numerical Value
- For each question, enter the correct numerical value of the answer using the mouse and the on-screen virtual numerical keypad in the place designated to enter the answer. If the numerical value has more than two decimal places, round off the value to TWO decimal places.
- The Marking scheme to evaluate Answer to each question will be :
Full Marks: +3 (If ONLY the correct numerical value is entered)
Zero Marks: 0 (In all other cases)

31 A gas obeys the equation : $P(V - nb) = nRT$. At what pressure (in atm) and 27°C , the density of gas becomes 8.0 g/L ? The molar mass of gas is 80 g/mol and $b = 0.4 \text{ L/mol}$.
($R = 0.08 \text{ L-atm/K-mol}$)



- 32 The vapour pressure of a pure liquid A is 70 torr at 27°C. It forms an ideal solution with another liquid B. The mole fraction of B is 0.2 and total vapour pressure of the solution is 84 torr at 27°C. The vapour pressure of pure liquid B at 27°C is $7x \times 10^1$. Then the value of x is
- 33 For the electrochemical cell,
 $\text{Mg(s)}|\text{Mg}^{2+}(\text{aq}, 1\text{M})||\text{Cu}^{2+}(\text{aq}, 1\text{M})|\text{Cu(s)}$
 the standard emf of the cell is 2.70 V at 300 K. When the concentration of Mg^{2+} is changed to x M, the cell potential changes to 2.67 V at 300 K. The value of x is _____.
 (given, $\frac{F}{R} = 11500 \text{ K V}^{-1}$, where F is the Faraday constant and R is the gas constant, $\ln(10) = 2.30$)
- 34 A mixture of ethanal and propanal is subjected to aldol condensation by treating with dilute NaOH. How many aldols (including stereo isomers) are expected to be formed in this reaction
- 35 The number of species those are more basic than $\text{C}_6\text{H}_5\text{NH}_2$



- 36 The number of ores that are subjected to Calcinations either for Concentration of ore or before the extraction of metal is
- | | | | |
|-----------|----------|-------------|-----------|
| Magnesite | Dolomite | Lime stone | Calamine |
| Malachite | Azurite | Bauxite | Celestite |
| Cerrucite | Cinnabar | Cassiterite | |

**MATHEMATICS****Max Marks: 62****SECTION – I
(SINGLE CORRECT ANSWER TYPE)**

This section contains 4 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which ONLY ONE option can be correct. **Marking scheme: +3 for correct answer, 0 if not attempted and –1 in all other cases. Section 1 (Max Marks: 12)**

- Section 1 contains Four questions
- Each Question has Four Options and Only One of these four will be the correct answer.
- For each question, choose the option corresponding to the correct answer
- The Marking scheme to evaluate Answer to each question will be :
Full Marks: +3 (If the answer is correct)
Zero Marks: 0 (If the question is unanswered)
Negative Marks: -1 (In all other cases)

37 If $A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ p & q & r \end{bmatrix}$ then $A^3 - rA^2 - qA =$

- A. pI B. qI C. rI D. PA

38 The area enclosed by the curve $y = \sqrt{4-x^2}$, $y \geq \sqrt{2} \sin\left(\frac{x\pi}{2\sqrt{2}}\right)$ and x-axis is divided by y-axis in the ratio is

- A. $\frac{\pi^2-8}{\pi^2+8}$ B. $\frac{\pi^2-4}{\pi^2+4}$ C. $\frac{\pi-4}{\pi+4}$ D. $\frac{2\pi^2}{2\pi+\pi^2-8}$

39 The number of solutions of the equation $z^2 + \bar{z} = 0$ is

- A. 1 B. 2 C. 3 D. 4

40 The range of values of a such that the angle θ between the pair of tangents drawn from $(a, 0)$ to the circle $x^2 + y^2 = 1$ lies in the interval $\left(\frac{\pi}{3}, \pi\right)$, is

- A. $(-2, -1) \cup (1, 2)$ B. $(-\sqrt{2}, 0) \cup (0, \sqrt{2})$ C. $(-\sqrt{3}, -\sqrt{2})$ D. $(-\sqrt{3}, -\sqrt{2}) \cup (\sqrt{2}, \sqrt{3})$

**SECTION – II
(ONE OR MORE CORRECT ANSWER TYPE)**

This section contains 8 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which ONE OR MORE than ONE option can be correct. **Marking scheme: +4 for all correct options & +1 partial marks, 0 if not attempted and -1 in all wrong cases**

Section 2 (Max Marks: 32)

- Section two contains Eight (8) questions
- Each Question has Four Options and One Or More Than One of these four will be the correct answer.
- For each question, choose the option(s) corresponding to all the correct answer(s)

The Marking scheme to evaluate Answer to each question will be :

Full Marks: +4 (If the answer is correct)

Partial Marks: +3 (If all the four options are correct but **only 3** are chosen)

Partial Marks: +2 (If THREE or More options are correct but **only 2** correct answers are chosen)

Partial Marks: +1 (If TWO or More options are correct but **only 1** correct answer is chosen)

Zero Marks: 0 (If the question is unanswered)

Negative Marks: -1 (In all other cases)



- 41 Let $f(x) = ax^2 + bx + c$; $a, b, c \in R$ and $a \neq 0$ suppose $f(x) > 0$ for all $x \in R$. Let $g(x) = f(x) + f'(x) + f''(x)$. Then
- A. $g(x) > 0 \forall x \in R$
- B. $g(x) < 0 \forall x \in R$
- C. $g(x) = 0$ has non real complex roots
- D. $g(x) = 0$ has real roots
- 42 Sides of $\triangle ABC$ are in A.P. If $a < \min\{b, c\}$ then $\cos A$ may be
- A. $\frac{3c-4b}{2b}$ B. $\frac{4c-3b}{2c}$ C. $\frac{4c-3b}{2b}$ D. $\frac{4b-3c}{2b}$
- 43 If the solution of $y \frac{d^2 y}{dx^2} + \left(\frac{dy}{dx}\right)^2 = x$, $y^2(0) = y^2(1) = 1$ is given by $y^2 = f(x)$ then:
- A. $f(x)$ is monotonically increasing $\forall x \in (1, \infty)$
- B. $f(x) = 0$ has only one root
- C. $f(x)$ is neither even nor odd
- D. $f(x)$ has 3 real roots
- 44 If A and B are invertible square matrices of the same order, then which of the following is correct?
- A. $\text{adj}(AB) = (\text{adj}B)(\text{adj}A)$
- B. $(\text{adj}A)' = (\text{adj}A')$
- C. $|\text{adj}A| = |A|^{n-1}$, where n is the order of matrix A
- D. $\text{adj}(\text{adj}B) = |B|^{n-2} B$, where n is the order of matrix B



- 45 Points A and B lie on the auxiliary circle of ellipse $\frac{x^2}{4} + y^2 = 1$. P and Q are the corresponding points on the ellipse for the points A and B respectively (O is the origin). Choose the correct option(s).
- A. The maximum value of angle AOP is $\tan^{-1} 2\sqrt{2}$
- B. The maximum value of angle AOP is $\tan^{-1} \frac{1}{2\sqrt{2}}$
- C. If $OA \perp OB$ and Q' is reflection of Q in origin, then minimum value of angle POQ' is $\tan^{-1} \frac{4}{3}$
- D. If $OA \perp OB$ and Q' is reflection of Q in origin, then minimum value of angle POQ' is $\tan^{-1} \left(\frac{3}{4} \right)$
- 46 Let $f(x) = \cos x \sin 2x$ then
- A. $\min f(x) > -7/9, x \in [-\pi, \pi]$
- B. $\min f(x) > -9/7, x \in [-\pi, \pi]$
- C. $\min f(x) > -1/8, x \in [-\pi, \pi]$
- D. $\min f(x) > -2/9, x \in [-\pi, \pi]$
- 47 Let $\vec{a}$ and $\vec{b}$ be two non-zero and non-collinear vectors then which of the following is/are always correct?
- A. $\vec{a} \times \vec{b} = [\vec{a}\vec{b}\hat{i}] \hat{i} + [\vec{a}\vec{b}\hat{j}] \hat{j} + [\vec{a}\vec{b}\hat{k}] \hat{k}$
- B. $\vec{a} \cdot \vec{b} = (\vec{a} \cdot \hat{i})(\vec{b} \cdot \hat{i}) + (\vec{a} \cdot \hat{j})(\vec{b} \cdot \hat{j}) + (\vec{a} \cdot \hat{k})(\vec{b} \cdot \hat{k})$
- C. If $\vec{u} = \hat{a} - (\hat{a} \cdot \hat{b}) \hat{b}$, $\vec{v} = \hat{a} \times \hat{b}$ and $|\vec{b}| = 1$ then $|\vec{u}| = |\vec{v}|$
- D. If $\vec{c} = \vec{a} \times (\vec{a} \times \vec{b})$ and $\vec{d} = \vec{b} \times (\vec{a} \times \vec{b})$ then $\vec{c} + \vec{d} = \vec{O}$
- 48 A bag initially contains one red and blue two balls. An experiment consisting of selecting a ball at random, noting its colour and replacing it together with an additional ball of the same colour. If three such trials are made, then:
- A. probability that atleast one blue ball is drawn is 0.9
- B. probability that exactly one blue ball is drawn is 0.2
- C. probability that all the drawn balls are red given that all the drawn balls are of same colour is 0.2
- D. probability that atleast one red ball is drawn is 0.6

**SECTION – III**
(NUMERICAL VALUE TYPE)

This section contains 6 questions. Each question is numerical value type. For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to second decimal place. (e.g. 003.2343, 6.250023, 7.000000, -0.33, -.30, 30.27000, -127.0030).

Marking scheme: +3 for correct answer, 0 if not attempted and 0 in all other cases. Section 3 (Max Marks: 18)

- This Section contains **Six (06)** Questions. The answer to each question is a Numerical Value
- For each question, enter the correct numerical value of the answer using the mouse and the on-screen virtual numerical keypad in the place designated to enter the answer. If the numerical value has more than two decimal places, **round off the value to TWO decimal places.**
- The Marking scheme to evaluate Answer to each question will be :
Full Marks: +3 (If ONLY the correct numerical value is entered)
Zero Marks: 0 (In all other cases)

- 49** The probability of an event A occurring is 0.5 and of B occurring is 0.3. If A and B are mutually exclusive events then the probability of neither A nor B occurring is
- 50** If the volume of a tetrahedron formed by the three coordinate planes and a variable plane is always 4 cubic units, then the locus of the foot of the perpendicular from the origin to this variable plane is $(x^2 + y^2 + z^2)^3 = kxyz$, where $k =$
- 51** If $x \in \left(0, \frac{\pi}{2}\right)$, one solution of $\frac{\sqrt{3}-1}{\sin x} + \frac{\sqrt{3}+1}{\cos x} = 4\sqrt{2}$ is $\frac{\pi}{12}$, the other solution must be $\frac{\lambda\pi}{36}$, then λ is
- 52** If the circle $x^2 + y^2 + 4x + 22y + c = 0$ bisects the circumference of the circle $x^2 + y^2 - 2x + 8y - d = 0$ then $c + d =$
- 53** Set A has 3 elements, set B has 4 elements. The number of surjections that can be defined from A to B is
- 54** If $\lim_{x \rightarrow 0} \left(1 + x \left(1 + \frac{f(x)}{kx^2}\right)\right)^{\frac{1}{x}} = e^3$ and $f(4) = 64$ then $k =$ _____